

SPARTA PRAGUE RATING SYSTEM



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ATTACKING THREAT

Metric Explanation

This was the first metric that came to mind after analysing the given dataset. While teams may be able to keep possession of the ball, the most important thing is what they're doing with the ball possession: are they transitioning the ball into the final third and getting closer to scoring with each pass?

Calculation Thought Process

Thanks to the existing metric 'obv_for_net', which measures the difference in attacking threat posed by a team between each pass (considering ball progression, player positioning, etc.), I was able to create a metric that provides an overview of the attacking threat of each team as a score out of 100. This score represents the attacking effectiveness of different teams in the final third of the pitch. To create the value of an attacking threat, I combined five weighted components because I'm aware that attacking threat can be generated in different ways.

Components

TOTAL OFFENSIVE VALUE

35% WEIGHTING

CALCULATED BY THE TOTAL OBV_FOR_NET FOR A TEAM OVER THE COURSE OF A GAME, IN THE ATTACKING THIRD

PASS SUCCESS RATE

25% WEIGHTING

CALCULATION OF THE PERCENTAGE OF COMPLETED PASSES IN THE ATTACKING THIRD (MEASURED BY $(\text{COMPLETED PASSES} / \text{TOTAL PASSES}) * 100$)

POSITIVE OBV RATE

15% WEIGHTING

THE PERCENTAGE OF ATTACKING THIRD PASSES WITH A POSITIVE OBV OUT OF ALL THE TEAM'S ATTACKING THIRD PASSES WITHIN A GAME

AVERAGE XG PER SHOT

15% WEIGHTING

MEASURES THE AVERAGE QUALITY OF EACH SHOT TAKEN, I.E. THE PROBABILITY OF A SHOT TAKEN BY THE TEAM RESULTING IN A GOAL

AVERAGE OBV PER PASS

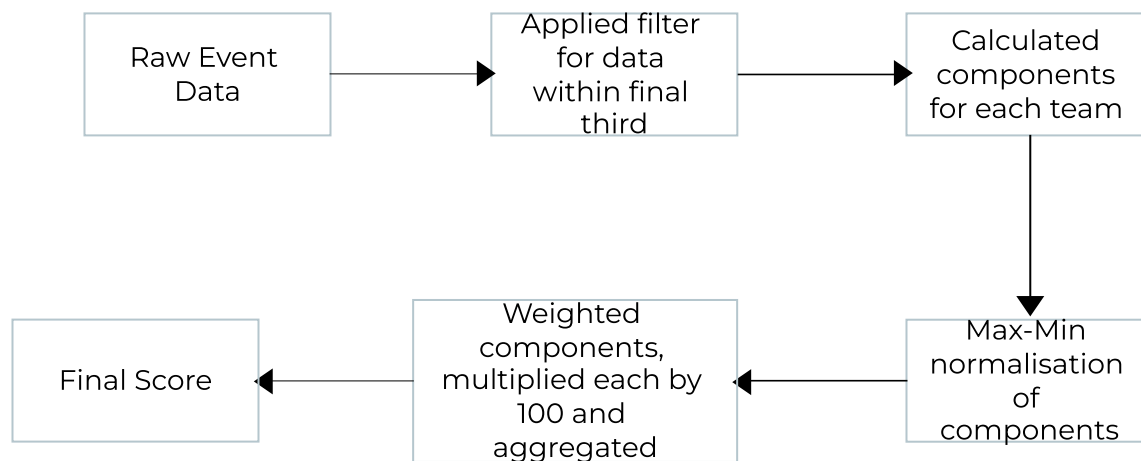
10% WEIGHTING

THE AVERAGE OFFENSIVE VALUE GENERATED PER PASS.
CALCULATED BY THE TEAM'S TOTAL OBV_FOR_NET / TOTAL PASSES

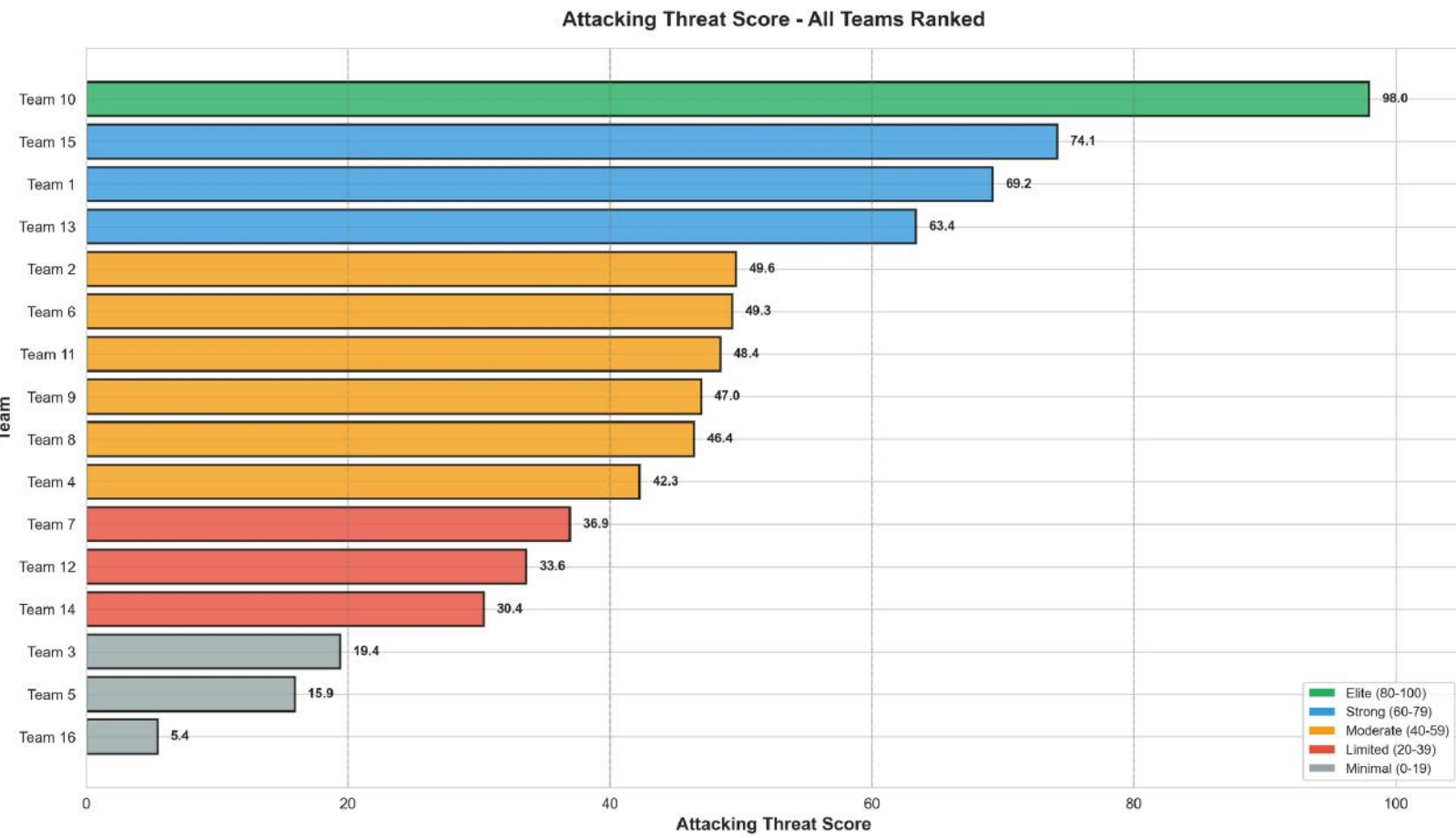
NORMALISATION OF COMPONENTS



These components were normalised (placed between 0 and 1, with 0 representing the minimum and 1 representing the maximum) before I assigned them their respective weightings. This is because they aren't all in the same format (some percentages, some decimals). Without normalisation, larger values like a total obv of 31.5 and a success rate of 73% would dominate smaller values like a 0.0064 average obv per pass. This flowchart shows the calculation process



RESULTS



N.B: Team 10's attacking threat value is made up of a total OBV of 31.5 (1st), a pass completion success rate of 73% (2nd), their average xG per shot is 0.122 (1st), an average OBV per pass of 0.0064 (1st), and they have a positive OBV rate of 48.7% (3rd) in the final third.

Also, scores represent relative rankings within the given dataset, not absolute performance percentages. A score of 98/100 means Team 10 comfortably ranks highest among these teams, not that they perform at a 98% capacity

Findings

Team 10's attacking threat score of 98 makes them extremely dangerous in attack, with over double the average score of 45.6. This data shows they excel in the attacking phases of the game, with the ability to progress the ball well enough to create a chance or pose goalscoring threat to the opposing team at a high rate.

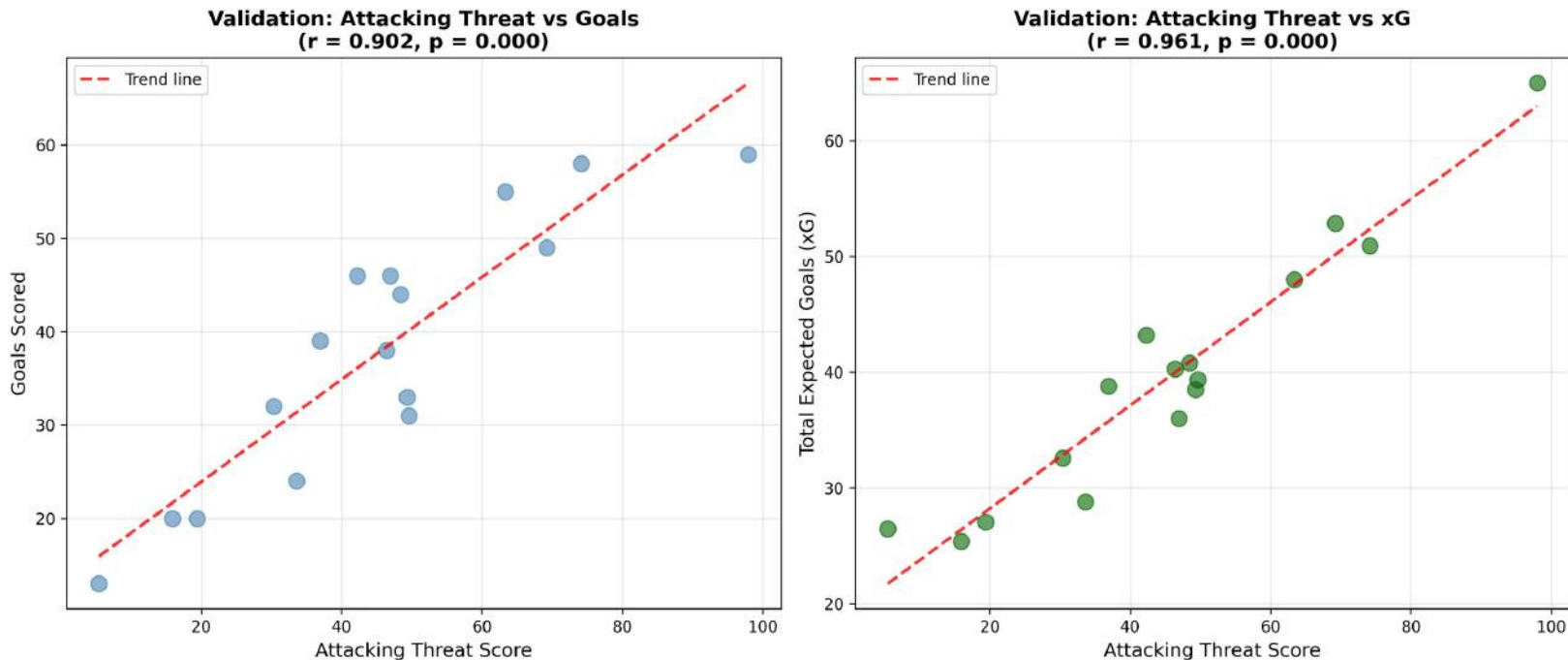
Teams 15, 1 and 13 are teams that are strong in attack, with an attacking threat also above average, but these teams are just below the standard of an elite attacking team.

Teams 2, 6, 11, 9, 8 and 4 are the subset of teams that have a standard level attack. The attacking threat of these teams can neither be classed as great nor poor. These teams are able to catch defenses out but at a moderate rate.

Team 16's 5.4/100 means they hold very little substance in attack and are not threatening at all in comparison to other teams. This data could suggest to their manager that they may need a change in their attacking philosophy to compete.

USABILITY OF METRIC AND LIMITATIONS

Usability



These diagrams show the correlation between the attacking threat score of a team and the overall goals they've scored, as well as the correlation between their attacking threat and their xG. The r coefficients of 0.902 and 0.961 mean there's a very strong correlation, proving the metric of attacking threat to be accurate.

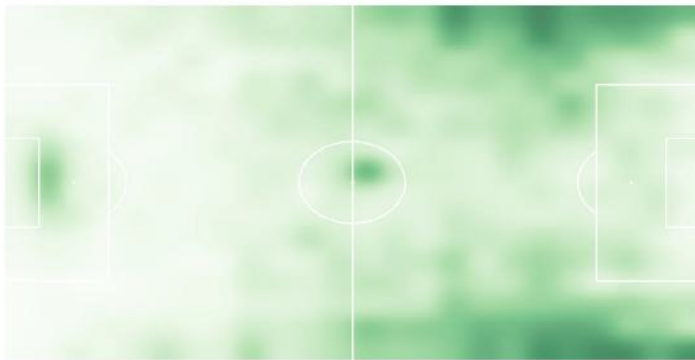
Limitations

One way to improve this data is to consider pass statistics within the middle and defensive thirds for teams that play a counterattacking style of football. For these types of teams, which operate at a higher tempo, attacks may often begin from their defensive or middle third, making them a significant attacking threat not just in the final phase of the pitch. If the given dataset considered the playstyles of teams, this is something I would measure.

ATTACK ORIGINS - TEAM 10 INSIGHT

Why Attack Origins?

Once I was able to determine the attacking efficiency of Team 10, I became curious about how they generated their attacks. I wanted to know what area of the pitch their most threatening attacks were starting from. This could give opponent teams insight into potential danger they may need to be aware of going into the game. To do this, I used the data from the location column of the dataset, as well as the obv_for_net column, to gather the pitch location of the highest 25% of OBV actions, whether they were ball carries, dribbles, passes, or shots. Once I was able to filter this data, I created a high-value attack zone distribution grid.



KEY INSIGHTS - TEAM 10

1. DOMINANT HIGH-VALUE ZONE (Most Frequent):

Zone: Left - Attacking Third

Actions: 1136 (20.2% of high-value actions)

- This is where Team 10 most often creates danger

2. MOST VALUABLE ZONE (Highest Total OBV):

Zones: Center - Attacking Third

Total OBV: 44.971

Actions: 523

Avg OBV per action: 0.0860

- This zone generates the most offensive value overall

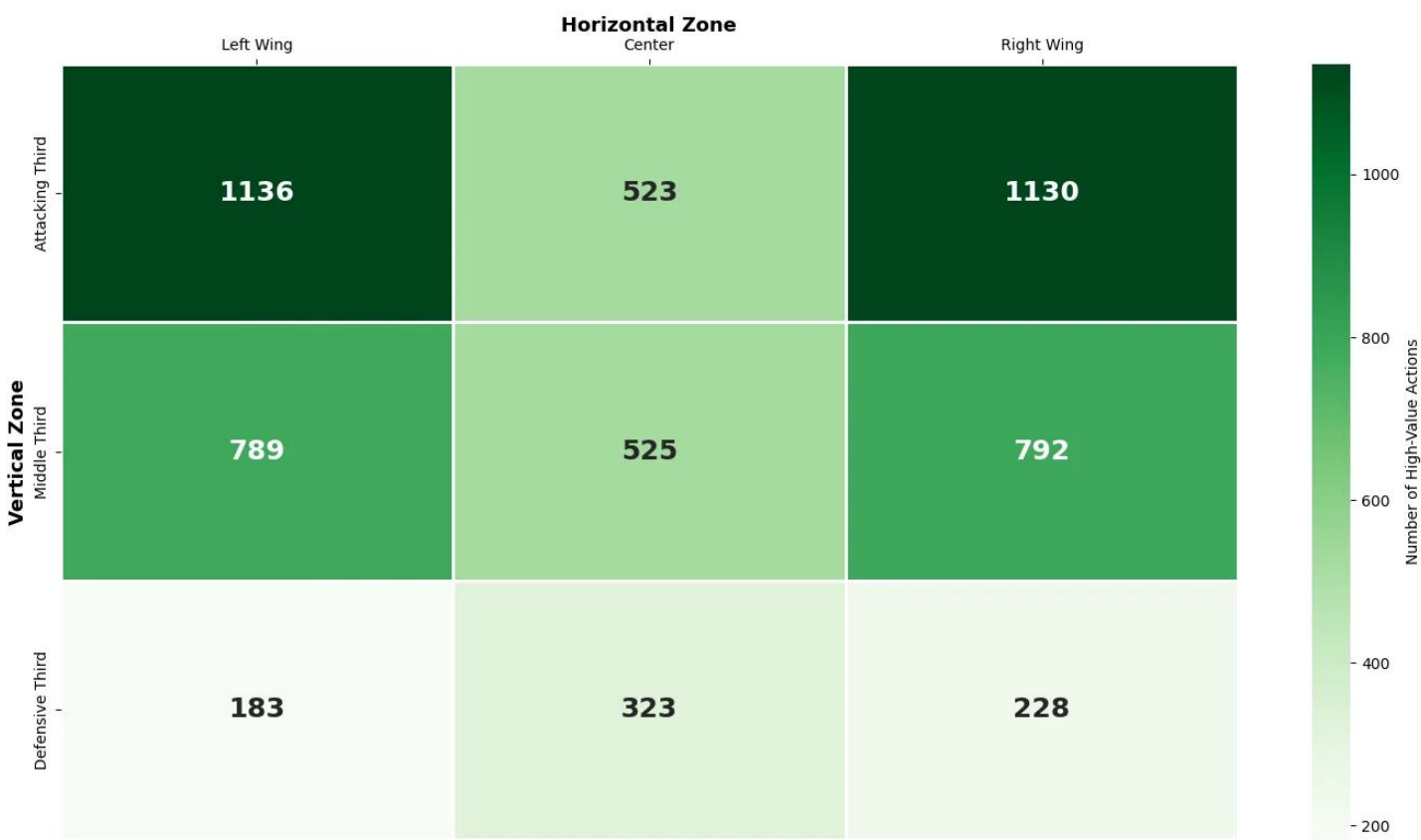
▲ INSIGHT:

• Most frequent danger: Left - Attacking Third

• Most total value: Center - Attacking Third

- Different zones for volume vs. value creation

Team 10: High-Value Zone Distribution
(Top 25% Most Impactful Actions)



This grid shows that Team 10 has a very balanced attack, with almost equal actions deployed on both flanks, indicating that they are just as dangerous on the right as they are on the left. This is a huge contributor to why they have an attacking threat score of 98; they have no attacking weakness. Although they have significantly fewer actions in the centre of the pitch in the attacking third, this is their most valuable attacking zone. The explanation for this is that the centre tends to be the tightest area of the pitch for all attacking teams as it's a heavily occupied space; however, this is the area that contains the most goal-scoring positions, hence the high average obv per action.

DEFENSIVE DISRUPTION

Metric Explanation

Another key aspect of the game that I wanted to measure is the defensive side, how disciplined teams are out of possession. While teams may be able to keep possession well, being able to regain the ball efficiently on the occasions they lose it is just as important. I wanted to compare the efficiency of different teams in not only regaining the ball but also being able to disrupt the attacking flow of the opposition. I used the following components to form this metric and then normalised them (See Page 4):

Components

CLEAR TURNOVER RATE

30% WEIGHTING

THE RATE OF DUELS THAT RESULTED IN A CLEAR POSSESSION TURNOVER

$(\text{NULL VALUES} / \text{TOTAL DUELS}) * 100$

LOOSE BALL WIN RATE

25% WEIGHTING

RATE OF DUELS THAT RESULTED IN LOOSE BALLS IN FAVOUR OF THE DEFENDING TEAM

$(\text{CALCULATED BY} - (\text{WON} + \text{SUCCESS IN PLAY} + \text{SUCCESS OUT} + \text{SUCCESS}) / (\text{WON} + \text{SUCCESS IN PLAY} + \text{SUCCESS OUT} + \text{SUCCESS} + \text{LOST IN PLAY} + \text{LOST OUT}))$

CLEAN RECOVERY RATE

20% WEIGHTING

A TEAM'S EFFICIENCY AT REGAINING POSSESSION WITHOUT FOULING

$[\text{TOTAL RECOVERIES} / (\text{TOTAL RECOVERIES} + \text{FOULS COMMITTED}) * 100]$

TOTAL RECOVERIES

15% WEIGHTING

$\text{INTERCEPTIONS} + \text{BALL RECOVERIES} + 50/50\text{S WON}$

HIGH IMPACT RATE

10% WEIGHTING

$\text{SUCCESSFUL DEFENSIVE ACTIONS WITH OBV IN THE TOP 25\% MOST NEGATIVE} / \text{TOTAL SUCCESSFUL DEFENSIVE ACTIONS}$

LOOSE BALL COMPONENT WEIGHTING

Explanation

To calculate the Loose Ball Win Rate, I decided to weigh duel outcomes in relevance to how favourable they are for the defensive team. I was able to do this thanks to my knowledge of each duel outcome's definition from the documentation provided, combined with OBV analysis from the dataset.

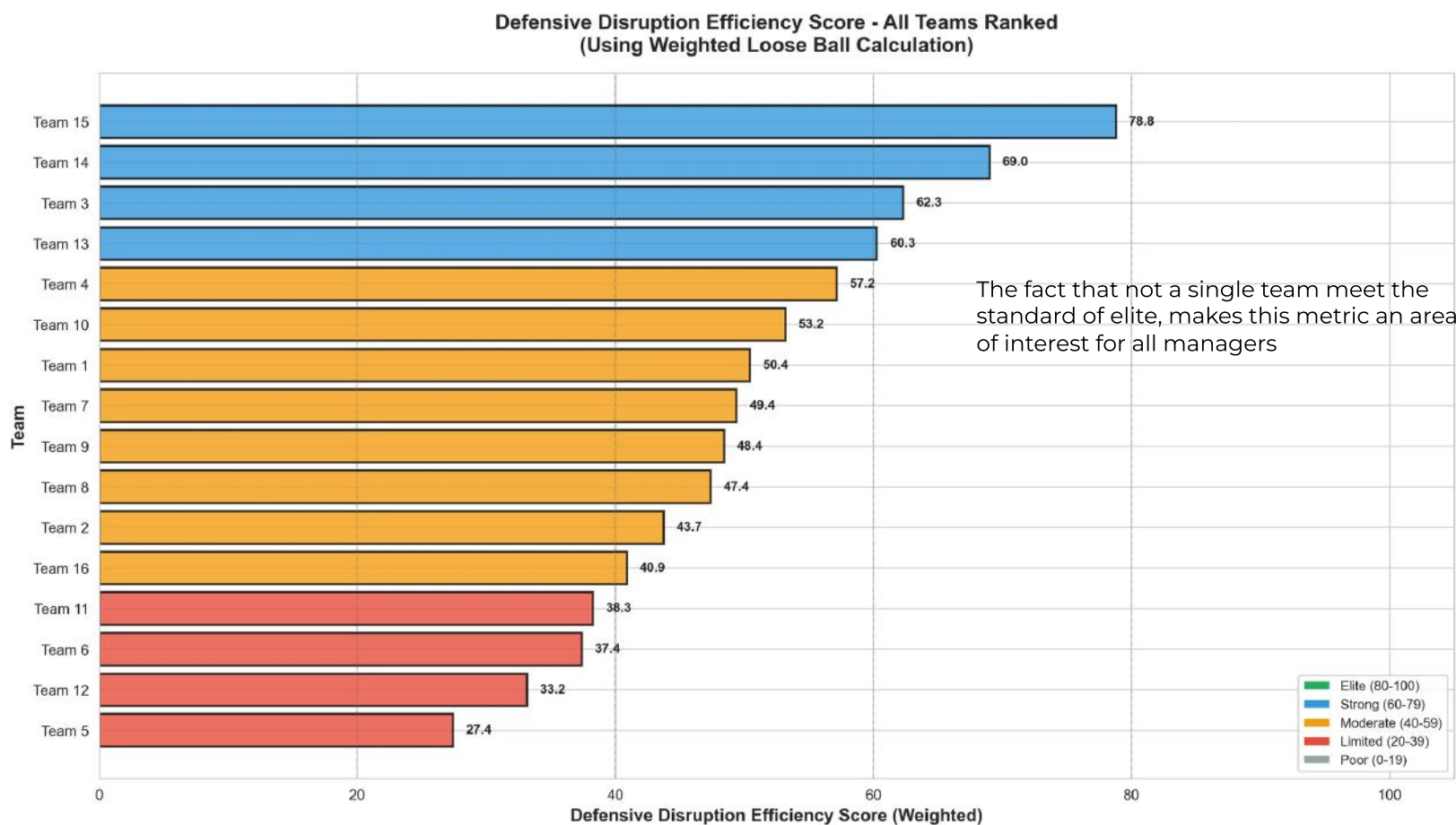
Weighting schema

Outcome	Weight	Rationale
Success In Play	*1.5	Ball knocked directly to teammate area. Highest OBV impact (-0.0136), best chance of maintaining possession
Success Out	*1.2	Ball knocked out of bounds in defender's favor, guaranteeing controlled restart (throw-in/corner)
Won	*1.0	Won physical contact, ball becomes contested. Baseline outcome - true 50/50 situation
Success	*1.0	Generic success outcome. Similar value to "Won" - contested situation

*This makes the formula Loose Ball Win Rate = $((\text{Won} * 1.0) + (\text{Success In Play} * 1.5) + (\text{Success Out} * 1.2) + (\text{Success} * 1.0)) / (\text{Won} + (\text{Success In Play} * 1.5) + ((\text{Success Out} * 1.2) + (\text{Success} * 1.0) + \text{Lost In Play} + \text{Lost Out}))$*

This weighted approach rewards teams that not only win physical contact battles but convert those battles into favourable positioning for their team. Teams that excel at directing balls to teammates (Success In Play) receive the appropriate recognition for this higher-value defensive action.

RESULTS



N.B. Team 15 have a 58.5% clear turnover rate (4th), a 65.2% loose ball win rate (2nd), an 88.3% clean recovery rate (1st) 1939 total recoveries (2nd), and a 25.3% high impact rate (1st)
Also, scores represent relative rankings within the given dataset, not absolute performance percentages. A score of 78.8/100 means Team 15 ranks highest among these teams, not that they perform at a 78.8% capacity

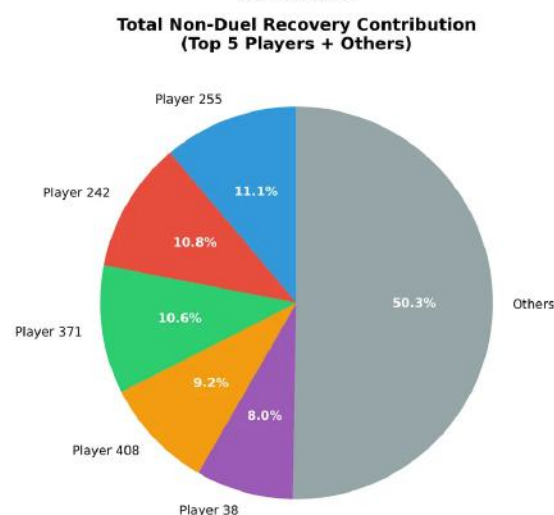
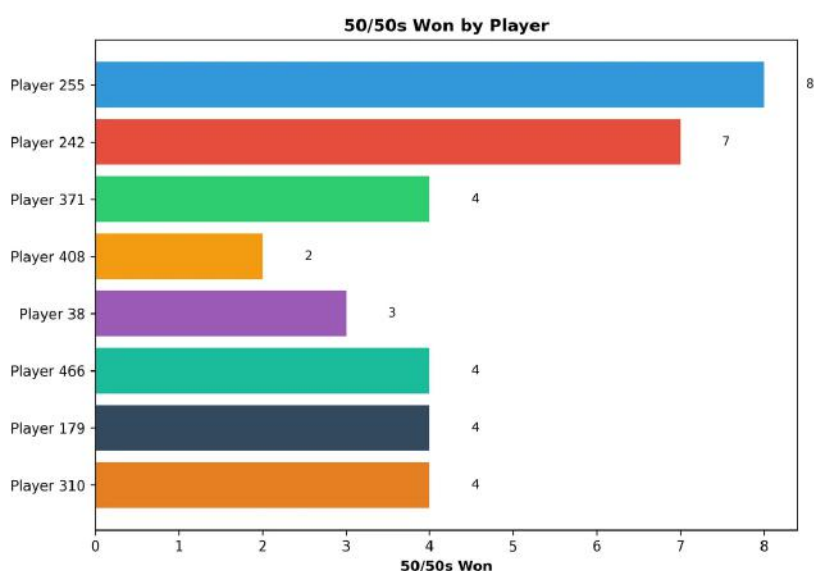
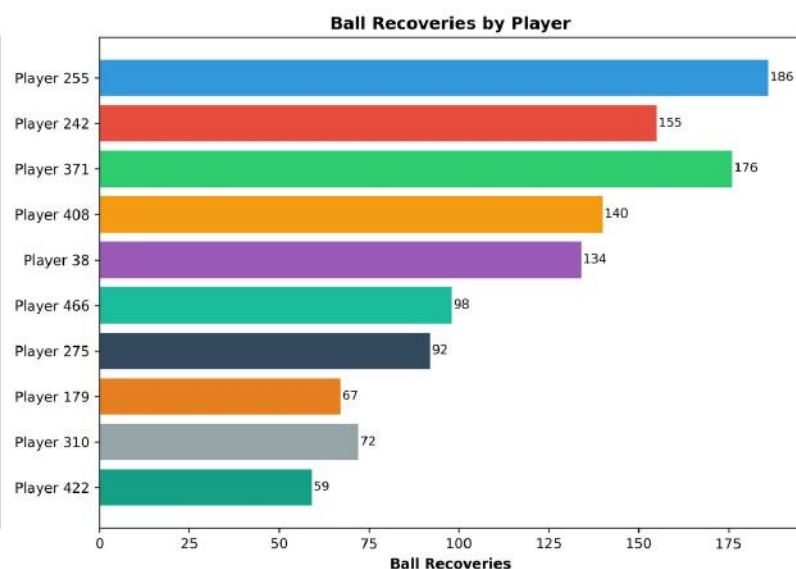
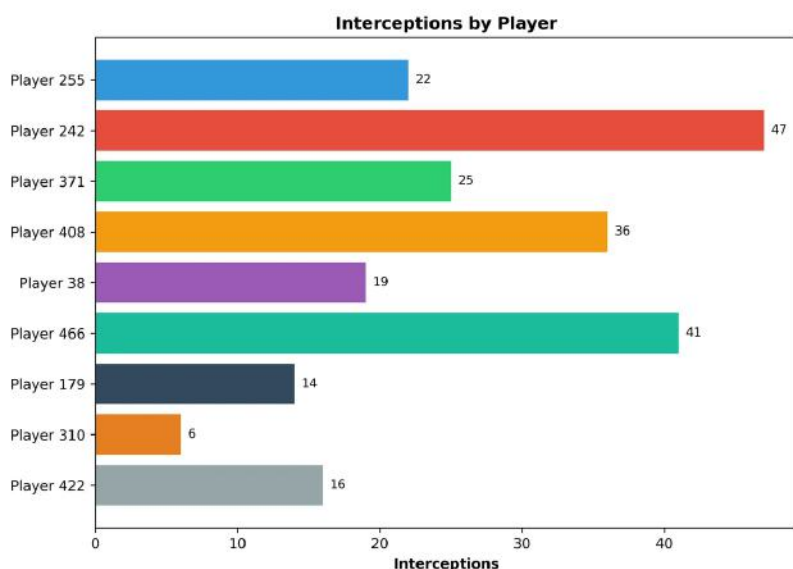
Findings

The data shows that Team 15 are the most efficient in disrupting the momentum of opponents, 78.8/100, far above the average of 49.8. This means that Team 15 excels at winning duels, recovering the ball and doing so without fouling. While Team 15 lead this category, Team 14's off-ball efforts are also worth taking note of, with 69/100. Team 3 and 13 also stand in the category of strong defensive teams.

Teams 4, 10, 1, 7, 9, 8, 2 and 16 all stand in the category of teams with moderate level off-ball capabilities, all being around the average defensive disruption levels. This means they can compete at a standard that matches the average level of attacking systems within these sixteen teams.

Teams 11, 6, 12 and 5 represent the teams that struggle to match the attacking intensity of their opponents while out of possession. A strong attacking team (from the Attacking Threat metric) are likely to be able to comfortably put on a display against these teams.

Team 15 - Player Recovery Contributions



The pie and bar charts above show the distribution of each player's contribution to total ball recoveries, both as a tally and as a percentage, for Team 15, the best off-the-ball team. This gives an insight into which player is most effective at winning the ball back in each aspect of winning the ball back. This data shows player 255 plays a huge factor in the off-ball successes of Team 15, with the most 50/50s won, the most ball recoveries and the most non-duel recoveries.

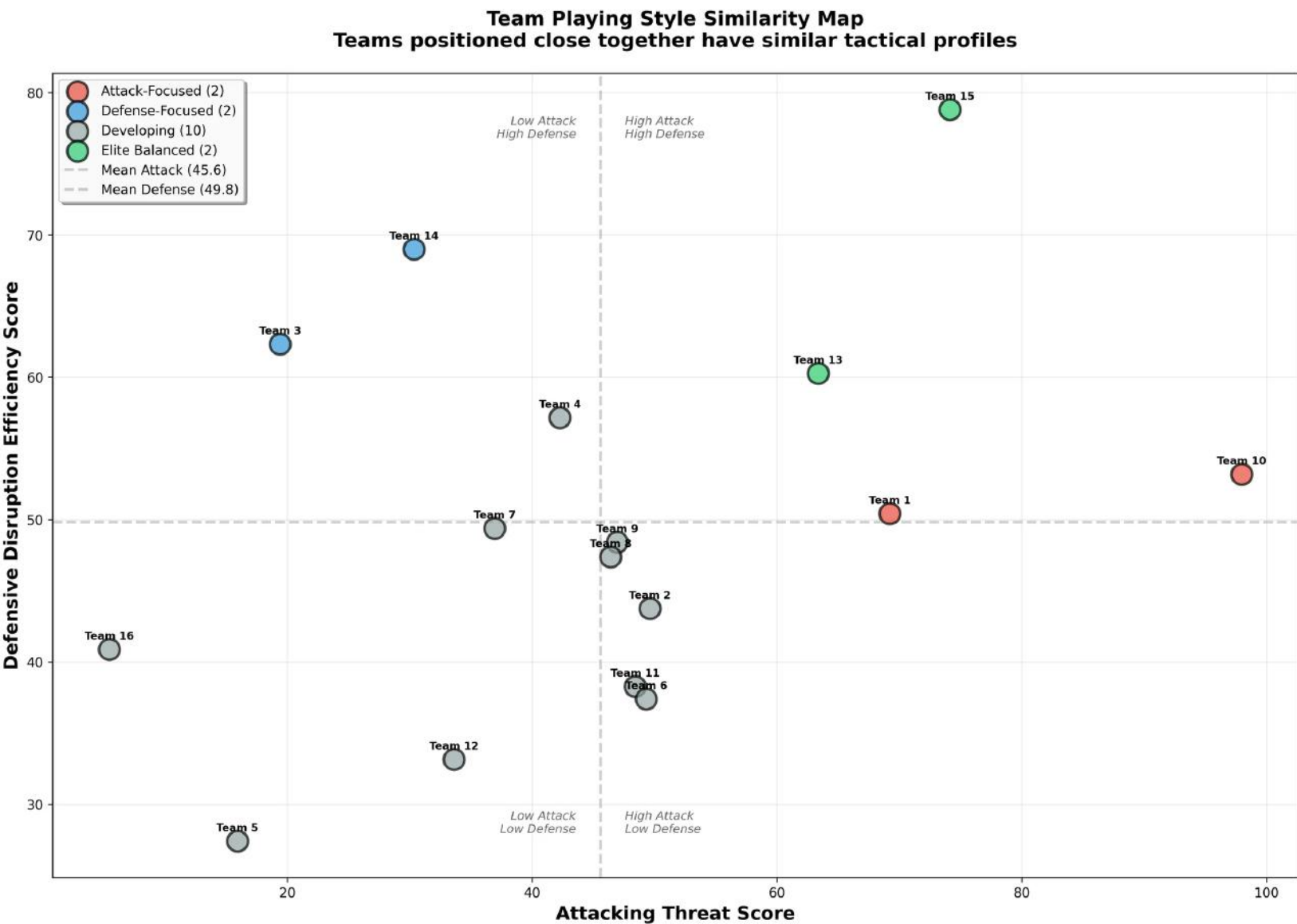
Usability

I wanted to visualise the negative correlation between defensive disruption efficiency and goals conceded, but due to the event-level data structure (without explicit match pairings), goals conceded per team could not be reliably calculated. However, the metric is validated through theoretical grounding; 'obv_against' is built on StatsBomb's validated defensive metrics.

Limitations

One issue with this metric is that it doesn't measure the distance covered by each team off the ball for them to have acquired their respective defensive disruption efficiency scores. This insight would help managers better understand the level of intensity and pressure required for their team to be able to keep opponents at bay, even when out of possession. If the dataset contained information about distances covered by players off the ball, this is certainly something I would have included in the metric.

SIMILARITY SYSTEM



This diagram puts the attack-defence balance of each team into perspective (based on our metrics). It shows which teams are similar in balance, which teams are more attack-oriented, which teams are more defence-oriented, as well as those that need development in both areas. From this diagram, I'd conclude that Team 15 are the most balanced team of the 16 teams. While Team 10, one could argue, whilst being a strong attacking force, are the most imbalanced, with an attack far above average but an average defence.

The majority of teams are in need of development in both areas of the game.

Teams positioned close together on the diagram are likely to result in even, competitive battles, while the managers of both teams will likely be encouraged to identify an area of the game where they can outperform their opponent.

Managers of teams in the attack-oriented and defence-oriented categories will look to use this in their favour to exploit opponents.

LIBRARIES USED

pandas, numpy, seaborn, matplotlib, scikit-learn, scipy, json,
warnings, pyarrow